

Microsoft Sentinel SOC Lab

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Introduction

For this project, I set up a small SOC environment using Microsoft Sentinel and a Windows virtual machine. The goal was to get familiar with collecting Windows security logs, writing KQL queries, investigating events, and documenting what I found.

The main things I investigated were:

- Failed login attempts
- Windows process creation
- PowerShell activity
- A file hash using VirusTotal

The Windows machine used for the lab was SOC-Windows01.

Incident 1: Failed Login Attempts

What I was investigating: I wanted to see how failed login attempts would appear in Microsoft Sentinel and what information I could get from the events. Windows Security Event ID 4625 is generated when a login attempt fails.

KQL Query

SecurityEvent

| where EventID == 4625

| project TimeGenerated, Computer, Account, IpAddress

| order by TimeGenerated desc

| take 20

Microsoft Sentinel | Logs

Service workspace: SOC-Lab-workspace

New Query 1

Time range: Last 24 hours | Show: 1000 results

1 SecurityEvent
 2 where: SecurityEvent:ID=1025
 3 project: TimeGenerated, Computer, Account, IpAddress
 4 order by: TimeGenerated desc | take 10

Results | Chart | Add bookmark

	TimeGenerated (UTC)	Computer	Account	IpAddress
☐	6/8/2024, 12:14:14 AM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 12:14:02 AM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 12:13:53 AM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 12:12:40 AM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 12:10:34 AM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 12:10:00 AM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 10:44:53 PM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 10:44:01 PM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 10:43:28 PM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150
☐	6/8/2024, 8:46:17:13 PM	SOC-Winserver01	SOC-Winserver01\psadmin	172.16.136.150

Showing 1-1 of 1 Display count

0s 442ms | Display time (UTC+00:00)

Query details

Figure 1. Failed login events shown in Microsoft Sentinel.

The results showed the time of the event, the computer, the account involved, and the IP address when that information was available. I also used a second query to see how many failed attempts occurred for each account.

SecurityEvent

| where EventID == 4625

| summarize FailedAttempts=count() by Account

| order by FailedAttempts desc



Figure 2. Number of failed authentication attempts by account.

Checking for a successful login. I also checked Event ID 4624 to see if there was a successful login after the failed attempts.

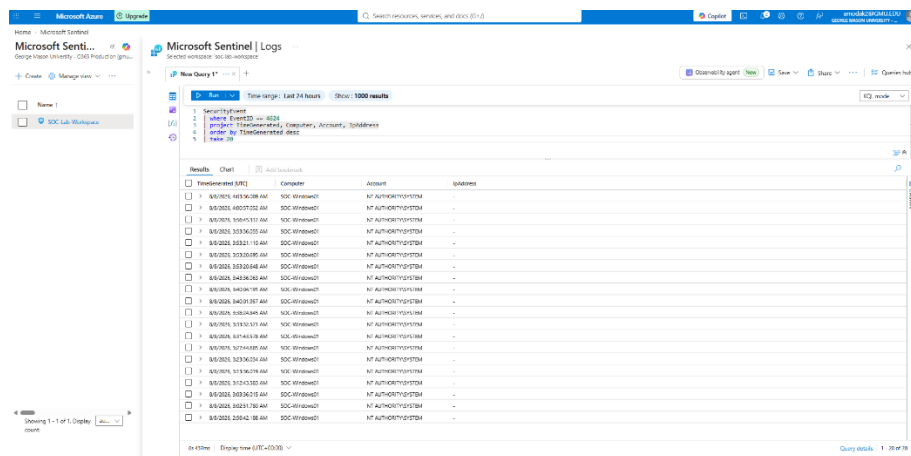
SecurityEvent

| where EventID == 4624

| project TimeGenerated, Computer, Account, IpAddress

| order by TimeGenerated desc

| take 20



The screenshot shows the Microsoft Sentinel interface with a query for EventID 4624. The query results are displayed in a table with the following columns: TimeGenerated, Computer, Account, and IpAddress. The results show 20 successful login events, ordered by TimeGenerated in descending order.

TimeGenerated	Computer	Account	IpAddress
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-
2023-05-05T10:00:00.000Z	DOC-Win10-01	NT AUTHORITY\SYSTEM	-

Figure 3. Successful login events checked during the investigation.

What I found: The failed login events showed that authentication failures were being collected successfully by Sentinel. Since this was my own lab environment, the activity was part of testing rather than an actual attack. In a real environment, I would pay more attention if there were many failures against one account, especially if a successful login happened shortly afterward.

MITRE ATT&CK

T1110 – Brute Force

If this happened on a production system, I would:

- Check with the user to see if the login attempts were expected.
- Look at the source IP address.
- Check for a successful login after the failures.
- Check MFA activity.
- Reset the password if I suspected the account was compromised.
- Revoke active sessions if necessary.
- Escalate the incident if the activity appeared malicious.

Incident 2 –PowerShell Process Investigation

What I was investigating: For this part, I wanted to see if I could find PowerShell activity in the Windows process creation logs. Windows Security Event ID **4688** records process creation.

I first tested the table with:

SecurityEvent

| where EventID == 4688

| take 20

This returned process creation events, so I used the SecurityEvent table for the rest of the investigation.

KQL Query

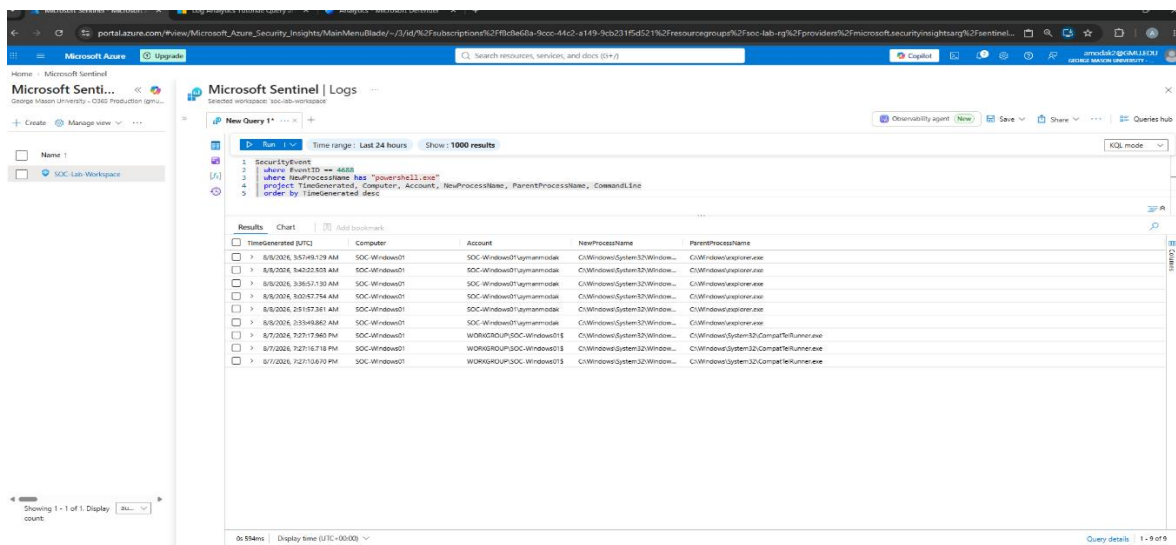
SecurityEvent

| where EventID == 4688

| where NewProcessName has "powershell.exe"

| project TimeGenerated, Computer, Account, NewProcessName, ParentProcessName, CommandLine

| order by TimeGenerated desc



The screenshot shows the Microsoft Sentinel interface with a KQL query executed. The query filters for SecurityEvent ID 4688 where the NewProcessName contains "powershell.exe". The results are ordered by TimeGenerated in descending order. The table displays columns for TimeGenerated, Computer, Account, NewProcessName, and ParentProcessName. The results show several PowerShell processes created by the SYSTEM account on SOC-Windows01, with parent processes being System32\WindowsPowerShell\powershell.exe. The last result shows a PowerShell process created by the WORKGROUP\SYSTEM account on SOC-Windows01, with a parent process being System32\CompattelRun.exe.

TimeGenerated	Computer	Account	NewProcessName	ParentProcessName
8/6/2024, 5:57:45.129 AM	SOC-Windows01	SOC-Windows01\SYSTEM	C:\Windows\System32\WindowsPowerShell\powershell.exe	C:\Windows\System32\WindowsPowerShell\powershell.exe
8/6/2024, 5:40:27.875 AM	SOC-Windows01	SOC-Windows01\SYSTEM	C:\Windows\System32\WindowsPowerShell\powershell.exe	C:\Windows\System32\WindowsPowerShell\powershell.exe
8/6/2024, 5:36:57.159 AM	SOC-Windows01	SOC-Windows01\SYSTEM	C:\Windows\System32\WindowsPowerShell\powershell.exe	C:\Windows\System32\WindowsPowerShell\powershell.exe
8/6/2024, 5:30:43.754 AM	SOC-Windows01	SOC-Windows01\SYSTEM	C:\Windows\System32\WindowsPowerShell\powershell.exe	C:\Windows\System32\WindowsPowerShell\powershell.exe
8/6/2024, 5:15:37.361 AM	SOC-Windows01	SOC-Windows01\SYSTEM	C:\Windows\System32\WindowsPowerShell\powershell.exe	C:\Windows\System32\WindowsPowerShell\powershell.exe
8/6/2024, 5:14:48.862 AM	SOC-Windows01	SOC-Windows01\SYSTEM	C:\Windows\System32\WindowsPowerShell\powershell.exe	C:\Windows\System32\WindowsPowerShell\powershell.exe
8/7/2024, 12:17:48 PM	SOC-Windows01	WORKGROUP\SYSTEM	C:\Windows\System32\CompattelRun.exe	C:\Windows\System32\CompattelRun.exe
8/7/2024, 12:17:18 PM	SOC-Windows01	WORKGROUP\SYSTEM	C:\Windows\System32\CompattelRun.exe	C:\Windows\System32\CompattelRun.exe
8/7/2024, 12:16:10 PM	SOC-Windows01	WORKGROUP\SYSTEM	C:\Windows\System32\CompattelRun.exe	C:\Windows\System32\CompattelRun.exe

Figure 4. PowerShell process creation events found in Microsoft Sentinel.

What I found:

One of the PowerShell events showed:

Computer: SOC-Windows01

User: SOC-Windows01\aymanmodak

Process:

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

Parent process:

C:\Windows\explorer.exe

Time:

8/8/2026 3:57:49 AM

The event showed that PowerShell was running under my lab user account and that Windows Explorer was the parent process. The command-line field was not populated in the data I collected, so I did not use it to make any assumptions about what the PowerShell process was doing.

PowerShell Test

To generate known PowerShell activity, I ran this command on the Windows VM: **Write-Host "SOC Lab PowerShell Test By Ayman Modak"** The PowerShell window returned: **SOC Lab PowerShell Test By Ayman Modak**. I was then able to see PowerShell process creation in Sentinel.

The PowerShell activity was expected because I generated it myself as part of the lab. If I found the same type of activity on a real workstation without knowing why it happened, I would investigate it further instead of immediately assuming it was malicious.

MITRE ATT&CK

T1059.001 – PowerShell

If the PowerShell execution looked suspicious, I would check:

- Who started PowerShell
- The parent process
- The command line
- Network connections
- Files created by PowerShell
- Other processes running around the same time
- Whether PowerShell was being used for persistence

If I confirmed malicious activity, I would consider isolating the computer and escalating the incident.

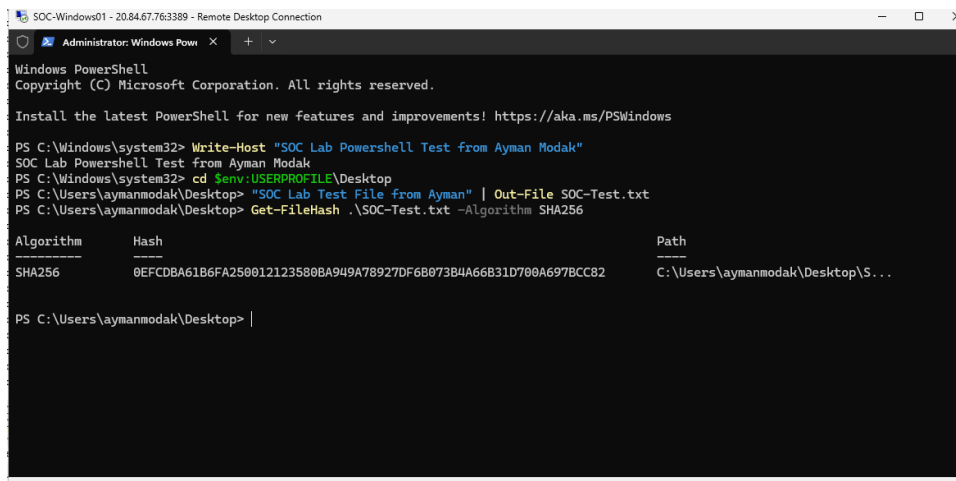
Incident 3 – File Hash Investigation

What I was investigating: For the last part of the lab, I wanted to practice taking a file, calculating its SHA256 hash, and checking that hash against a threat-intelligence service. I created a harmless text file called: **SOC-Test.txt**

The file was created specifically for this lab and did not contain personal or sensitive information.

Creating the hash

I used PowerShell: `Get-FileHash .\SOC-Test.txt -Algorithm SHA256`



```
SOC-Windows01 - 20.84.67.763389 - Remote Desktop Connection
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> Write-Host "SOC Lab Powershell Test from Ayman Modak"
SOC Lab Powershell Test from Ayman Modak
PS C:\Windows\system32> cd $env:USERPROFILE\Desktop
PS C:\Users\aymanmodak\Desktop> "SOC Lab Test File from Ayman" | Out-File SOC-Test.txt
PS C:\Users\aymanmodak\Desktop> Get-FileHash .\SOC-Test.txt -Algorithm SHA256

Algorithm      Hash                                                    Path
-----
SHA256         0EFCD8A61B6FA250012123580BA949A78927DF6B073B4A66B31D700A6978CC82  C:\Users\aymanmodak\Desktop\S...

PS C:\Users\aymanmodak\Desktop> |
```

Figure 5. SHA256 hash calculated for SOC-Test.txt.

SHA256 on VirusTotal

I searched the SHA256 hash on VirusTotal instead of uploading the file.

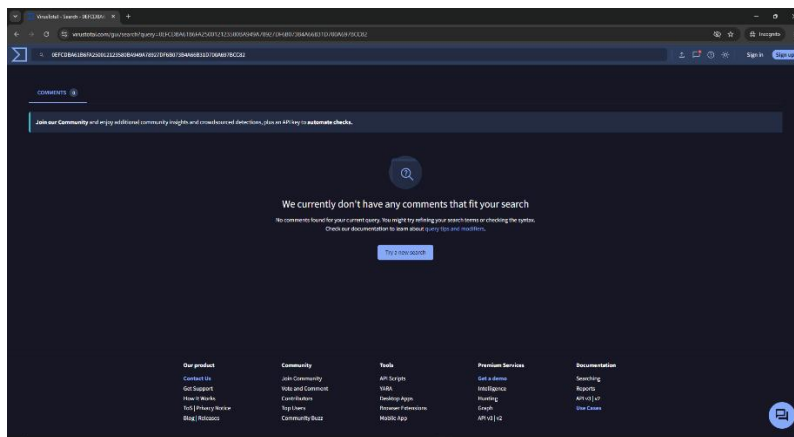


Figure 6. SHA256 hash checked using VirusTotal.

VirusTotal result: No existing VirusTotal analysis was found for the SHA256 hash. Since I created the file myself for the lab, this was expected.

This demonstrated a basic threat-intelligence workflow: **File** → **SHA256** → **VirusTotal** → **Reputation**

In a real investigation, a known malicious hash could be searched across an organization's endpoints to determine whether the same file exists elsewhere.

Overall Results

By the end of the lab, I was able to successfully collect and investigate Windows security events in Microsoft Sentinel.

I worked with:

Event ID 4625

Failed authentication

Event ID 4624

Successful authentication

Event ID 4688

Process creation

I also investigated PowerShell activity and used PowerShell to calculate a SHA256 hash for a test file.

What I Learned

One of the biggest things I learned from this project was that KQL queries cannot always just be copied from a tutorial and expected to work. For example, I initially expected to use a Sysmon table for process activity. Instead, I found that the process creation events I needed were available in the Security Event table using Event ID 4688.

I had to look at the data that was coming into Sentinel and adjust my queries based on the fields available. I also learned how different Windows event IDs can be used together. Event 4625 can show failed logins, Event 4624 can be used to check successful logins, and Event 4688 can show processes being created.

The PowerShell investigation also showed me that having more telemetry is useful. My current Event 4688 data showed the process, user, parent process, and timestamp, but the command-line information was not available. In a real SOC, having command-line data would make the investigation much more useful.

Skills Practiced

- Microsoft Sentinel
- KQL
- Windows Event Logs
- Event ID 4624
- Event ID 4625
- Event ID 4688
- PowerShell
- Process investigation
- Authentication investigation
- SHA256 hashing
- VirusTotal
- MITRE ATT&CK
- Basic incident response

Conclusion

This lab gave me hands-on experience using Microsoft Sentinel instead of only learning about SIEMs from a textbook or video.

I was able to collect Windows security events, write KQL queries, find failed authentication attempts, investigate PowerShell process creation, and perform a basic file-hash investigation.

The project also helped me understand that an SOC investigation is not just about finding an alert. The analysts must look at the available evidence, determine whether the activity makes sense, and decide what should happen next.

The next step for me would be to build more advanced detections and investigate additional types of suspicious activity.